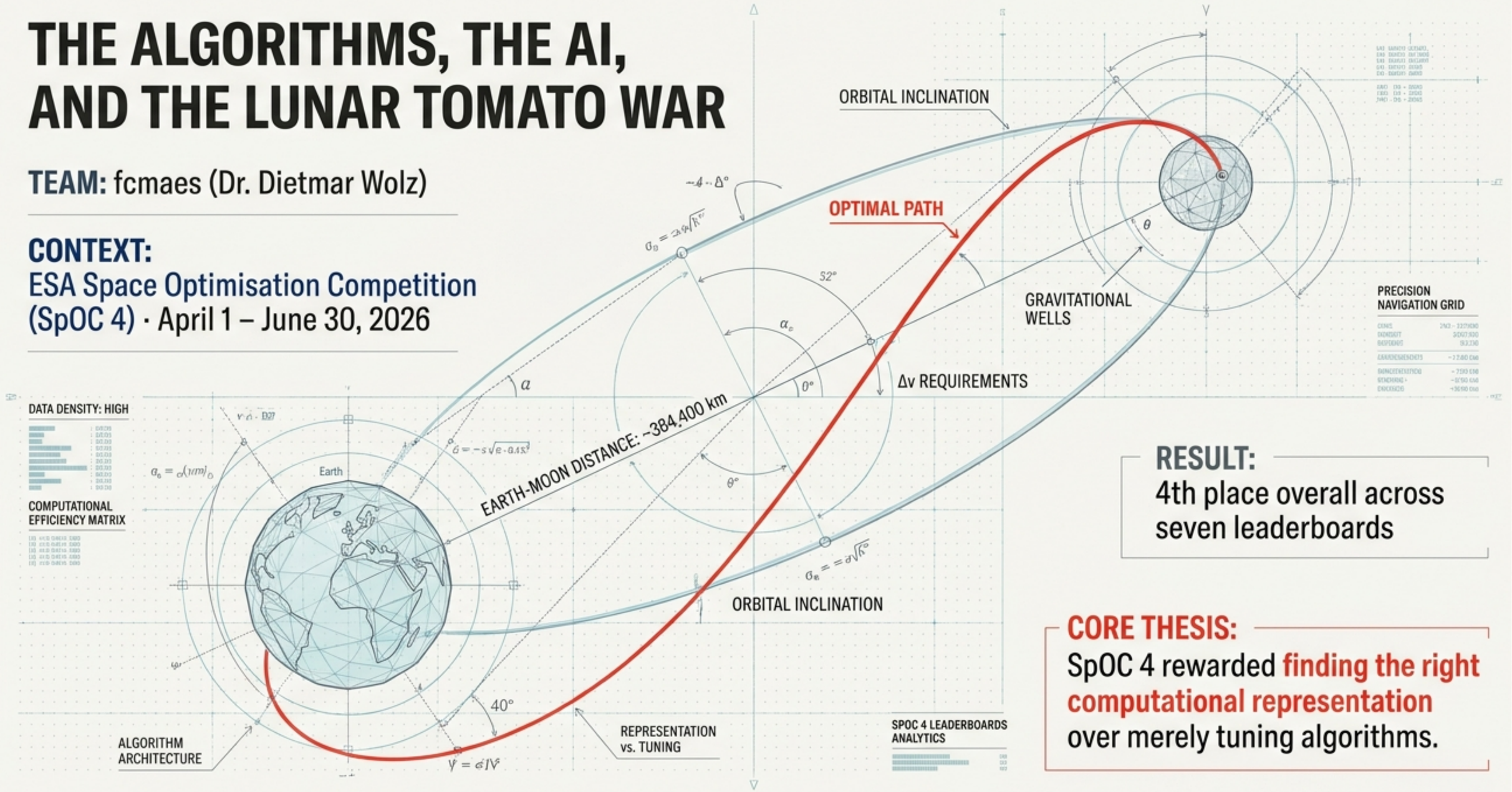
THE ALGORITHMS, THE AI, AND THE LUNAR TOMATO WAR

TEAM: fcmaes (Dr. Dietmar Wolz)

CONTEXT:
ESA Space Optimisation Competition (SpOC 4) · April 1 – June 30, 2026



RESULT:
4th place overall across seven leaderboards

CORE THESIS:
SpOC 4 rewarded **finding the right computational representation** over merely tuning algorithms.

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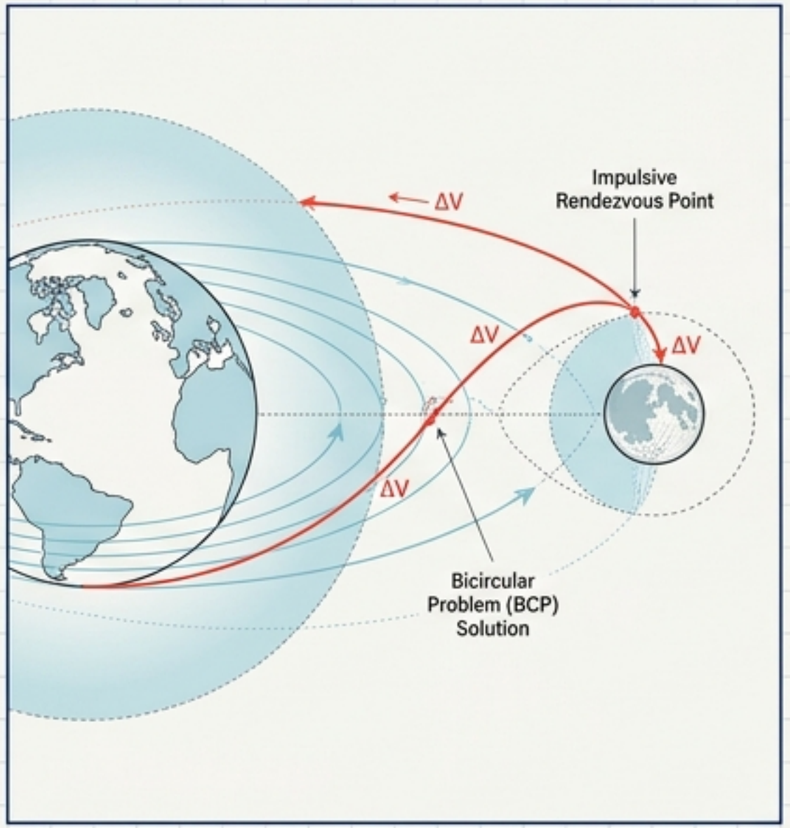
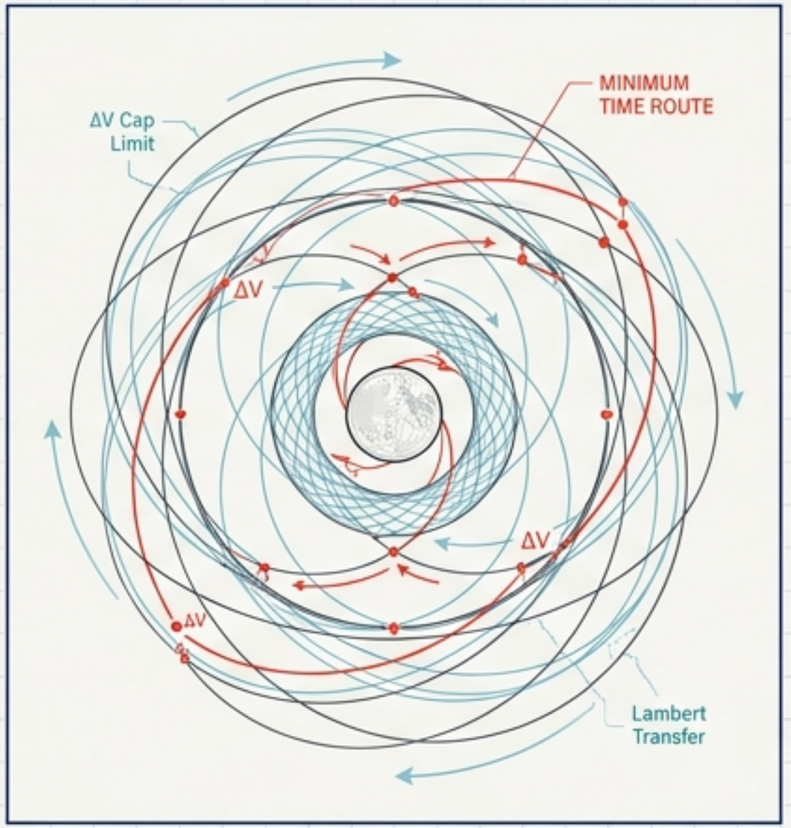
Three mathematically rigorous challenges disguised as lunar logistics



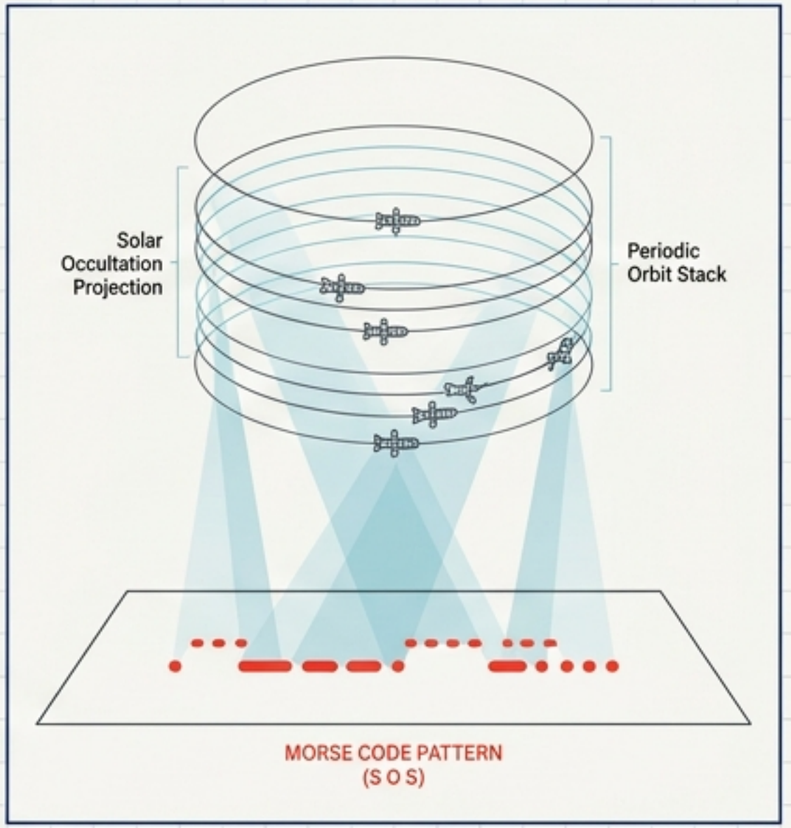
Challenge 1: Luna Tomato Logistics

Objective: Supply lunar outposts via Earth-Moon transfers. Requires 3-D matching and impulsive rendezvous design in the bicircular problem (BCP).



Challenge 2: Keplerian Tomato TSP

Objective: Route a single spacecraft to visit up to 1,051 lunar orbits in minimum time, governed by time-dependent Lambert transfers and strict ΔV caps.



Challenge 3: Luna Tomato Advertising

Objective: Configure spacecraft in periodic orbits so their solar occultations project Morse code onto the Moon.

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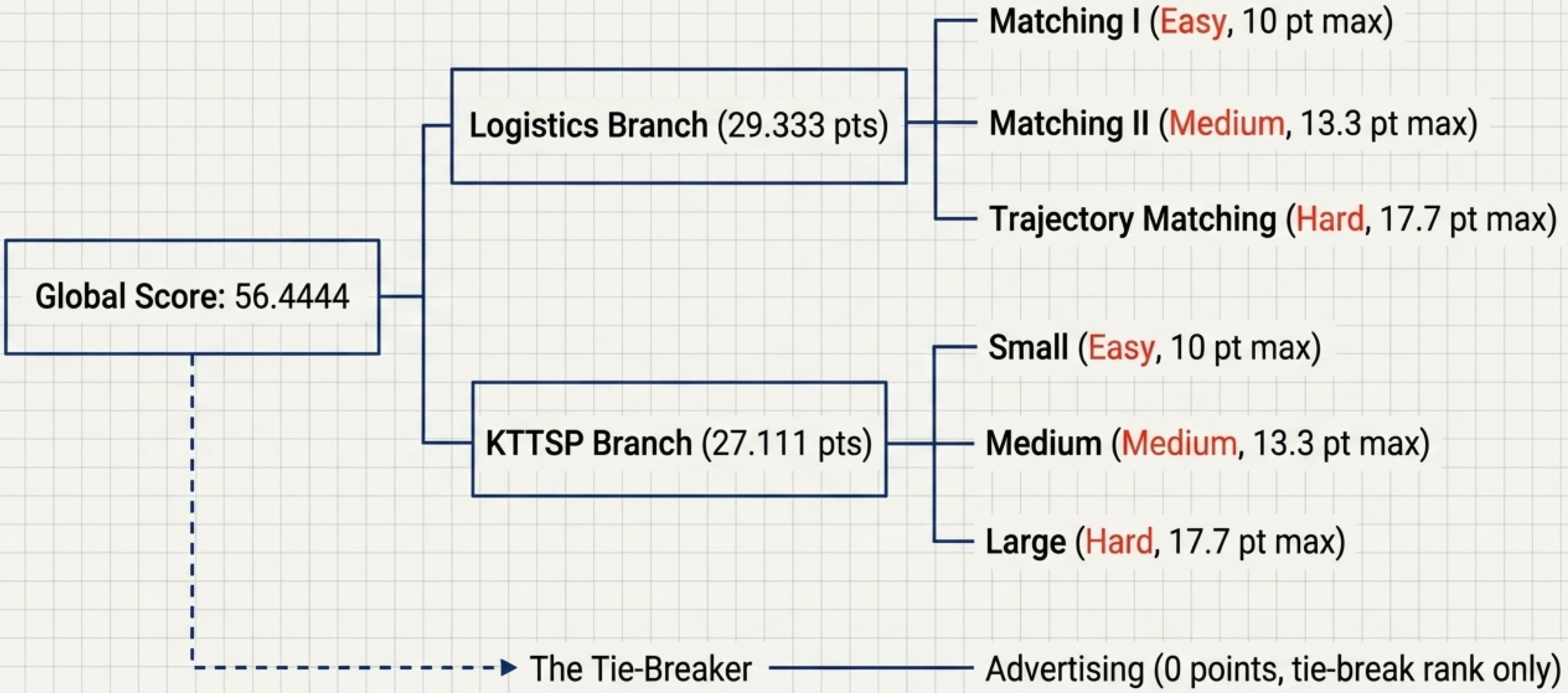
Six mandatory leaderboards drove the global score



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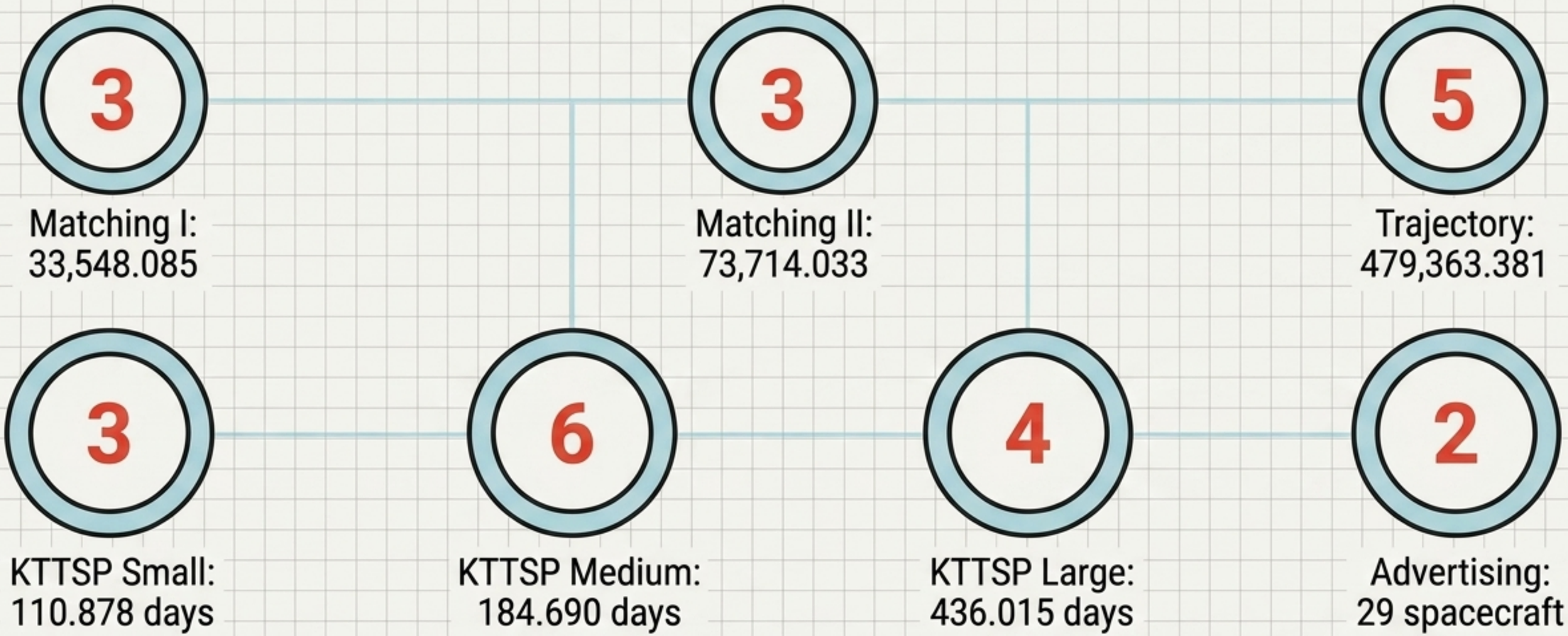
Fourth place achieved through continuous architectural pivots

Global Rank: 4 of 17 (56.4444 points)

Tie-Breaker Rank: 2



The Optimise platform minimizes, displaying logistics scores as negative. We report them here as positive delivered value.

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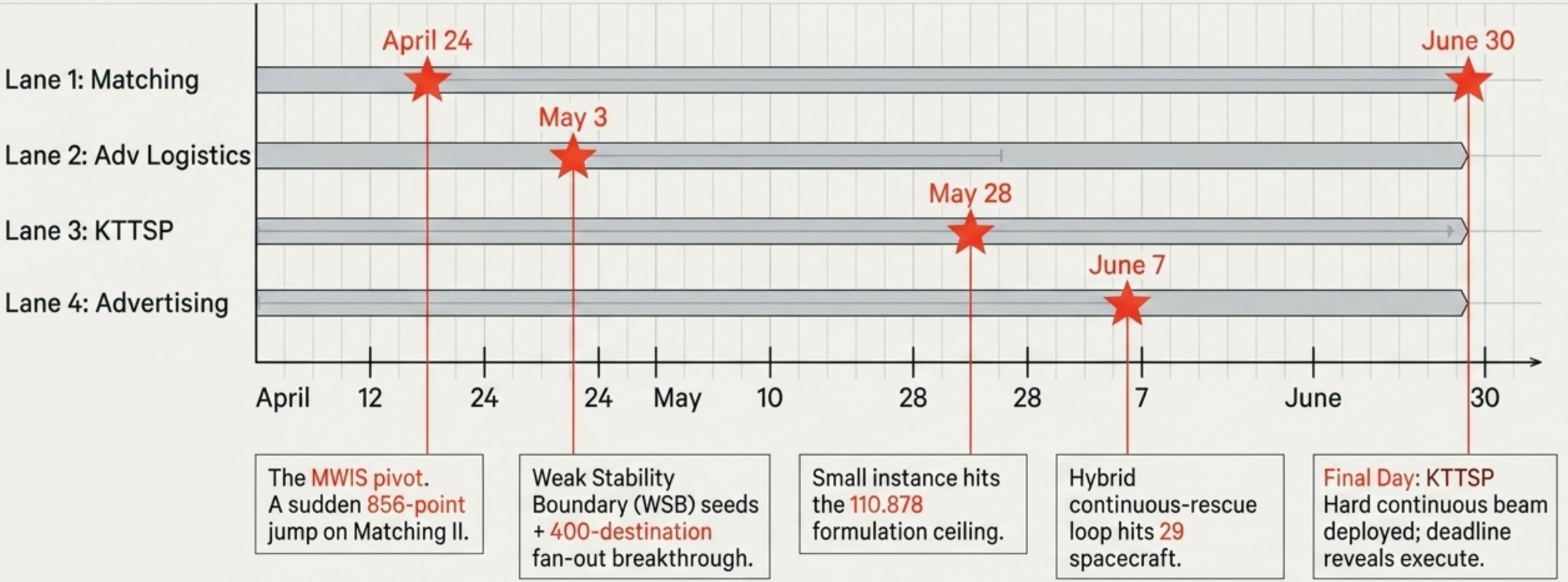
Four parallel method lanes and the timing of representation breakthroughs



Note: fcmaes led three boards until the final

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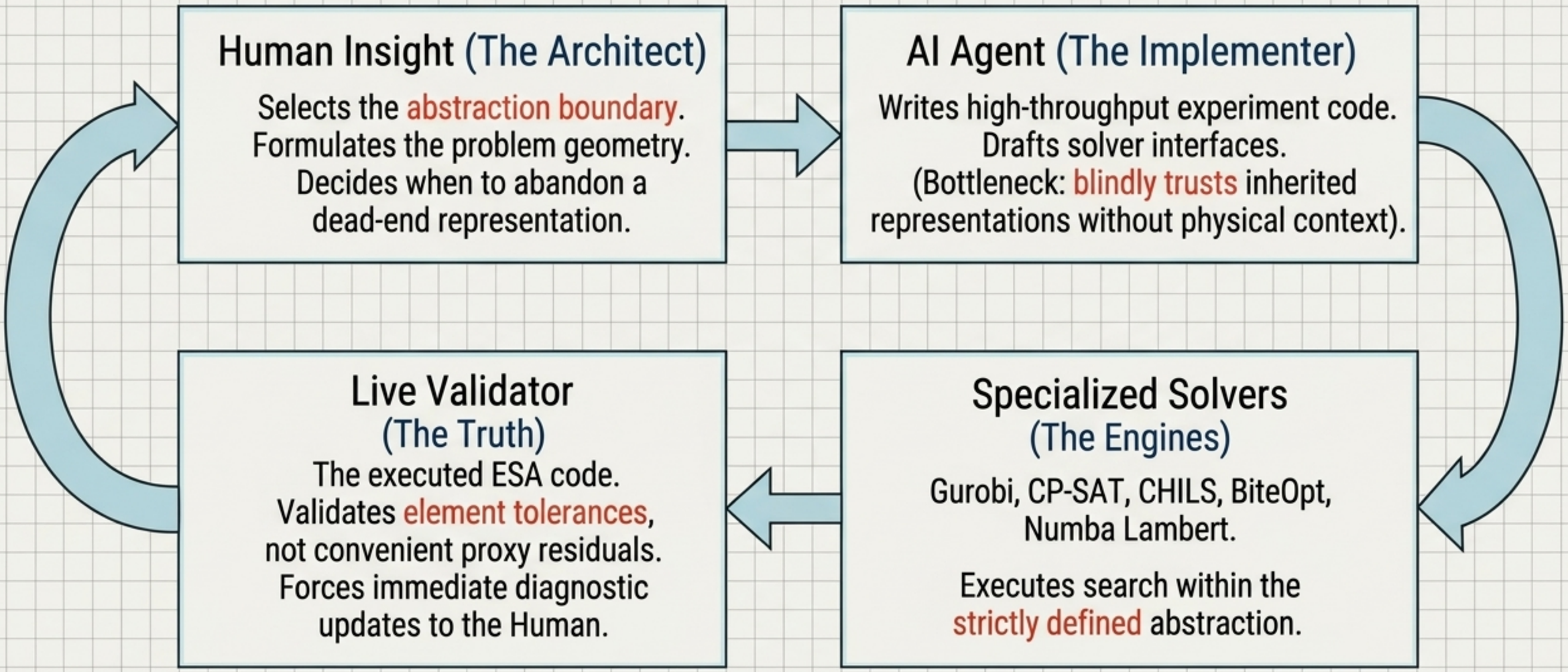
The campaign operating system: specialized roles, strict boundaries

Responsibility Matrix



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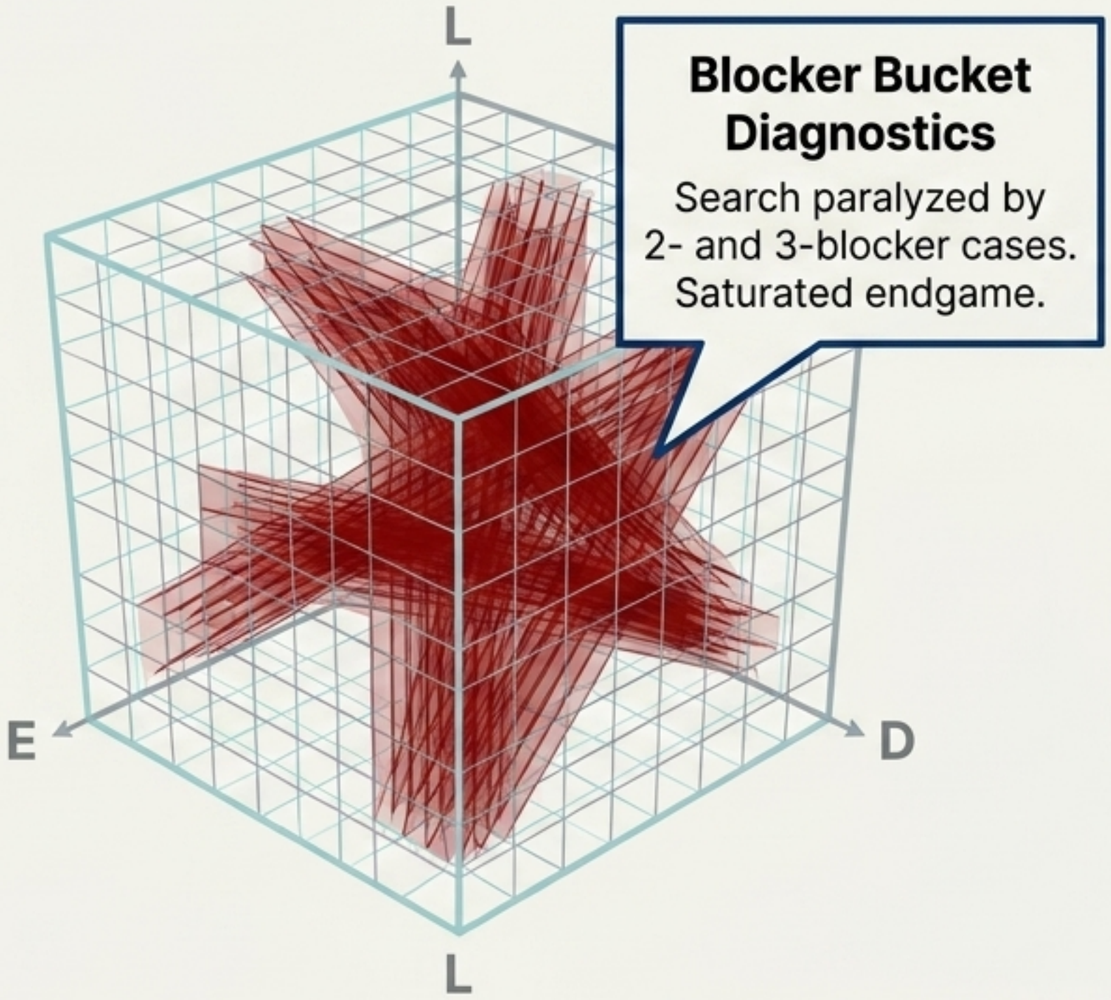
The 3-D Set Packing Trap



Blocker Bucket Diagnostics

Search paralyzed by 2- and 3-blocker cases. Saturated endgame.

The Task:	Select valuable (E, L, D) transfer triples without reusing any endpoint. Maximize sum of weights.
The Scale:	25,000 transfers (Matching I) and 92,103 transfers (Matching II).
The Trap:	Formulating this directly as an Integer Linear Program (ILP) with one binary per transfer.
The Result:	Local search and Gurobi polishing stalled near 72,713 on Matching II. Incumbent quality, not model correctness, became the bottleneck.

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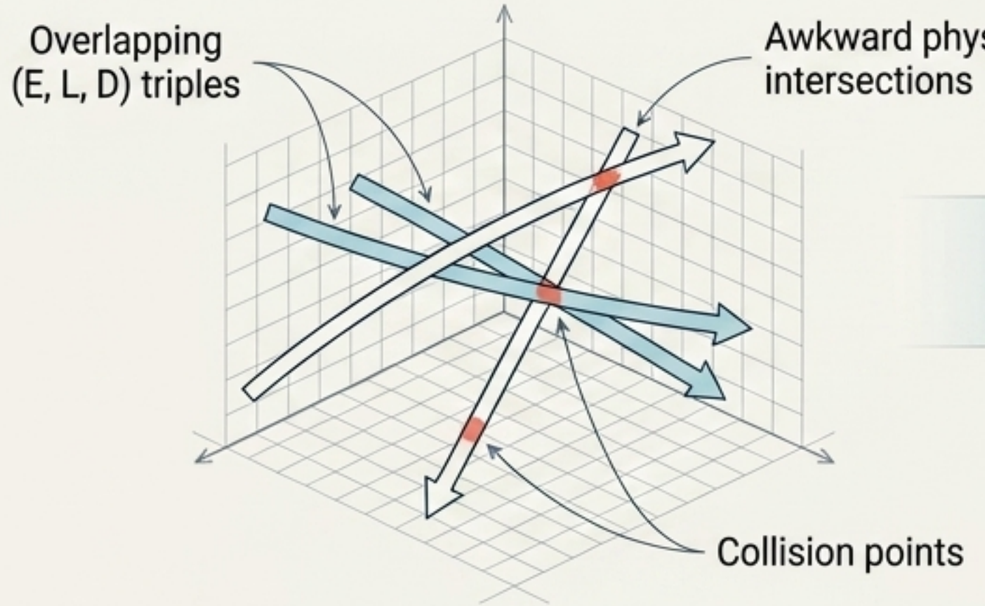
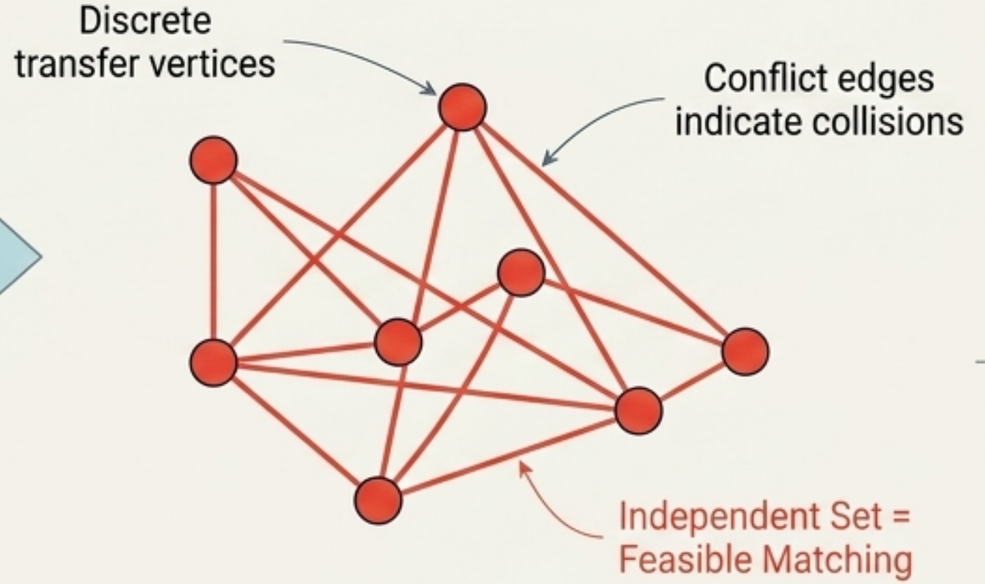
The decisive reformulation: Translating matching into MWIS

Physical Transfers (Matching Problem)



Reformulation

Mathematical Construct (MWIS)



The Reframing

Two vertices are adjacent if their triples collide. A feasible matching is exactly an independent set. Maximum weight matching becomes Maximum Weight Independent Set (MWIS).

The Engine

Patched CHILS
(a state-of-the-art MWIS solver).

The Catalyst

Cold-starting CHILS avoided anchoring to the old local optimum.

The Leap

The score jumped from 72,713 to 73,306 within a minute, eventually reaching the final 73,714.033.

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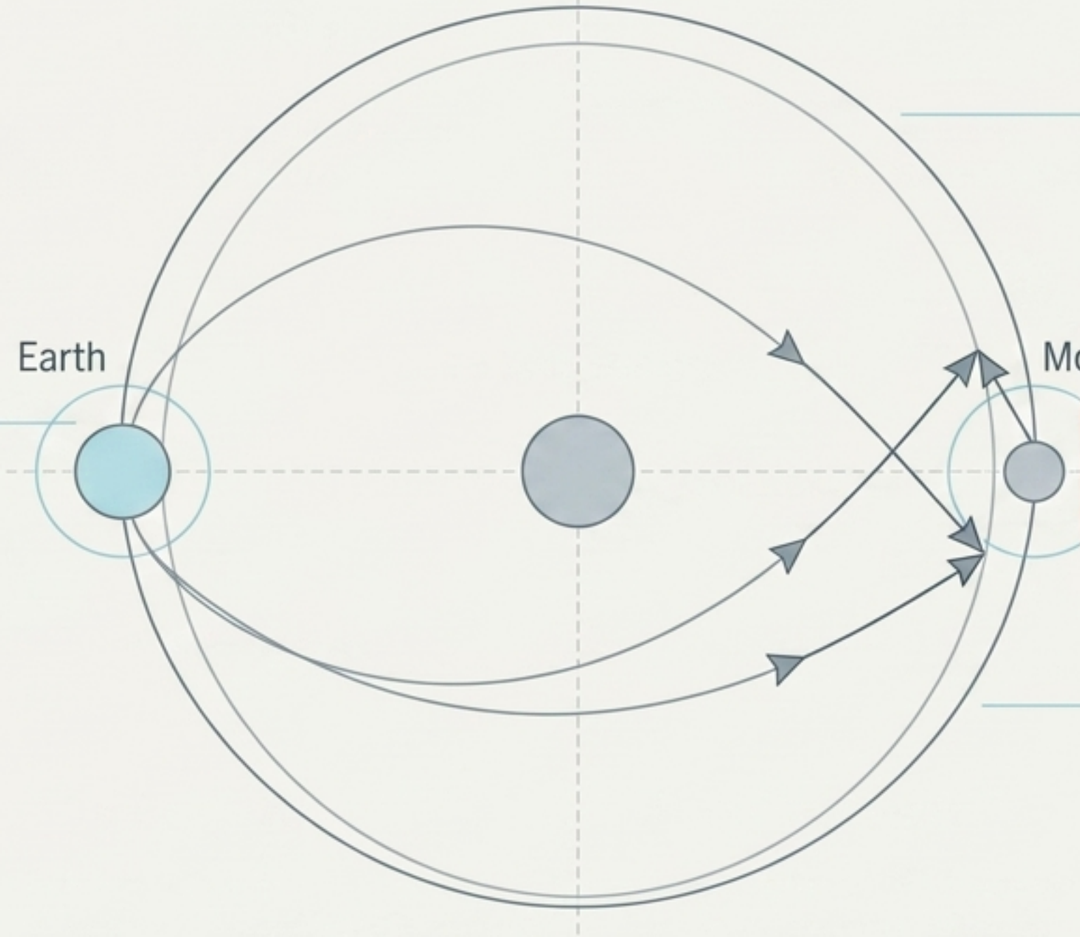
Advanced Trajectory Logistics: Generating the weights with physics

The Task:

Same 3-D matching rule, but the transfer weights must be computed via astrodynamics.



The Dynamics:

3-impulse rendezvous in the Earth-Moon-Sun Bicircular Problem.

Validator Conditions:
1e-6 tolerance on Keplerian elements (a, e, i)

The Chromosome:

400 x 21 = 8,400 values (timing, impulses, coordinates).

The Validator Forensics: Success wasn't a Cartesian coordinate match. The validator strictly checked a 1e-6 tolerance on Keplerian elements (a, e, i).

(Note: The uploaded validator implementation computed μ_s = 328900.559, exposing a 1.2% discrepancy with the README. We optimized against the executed truth.)

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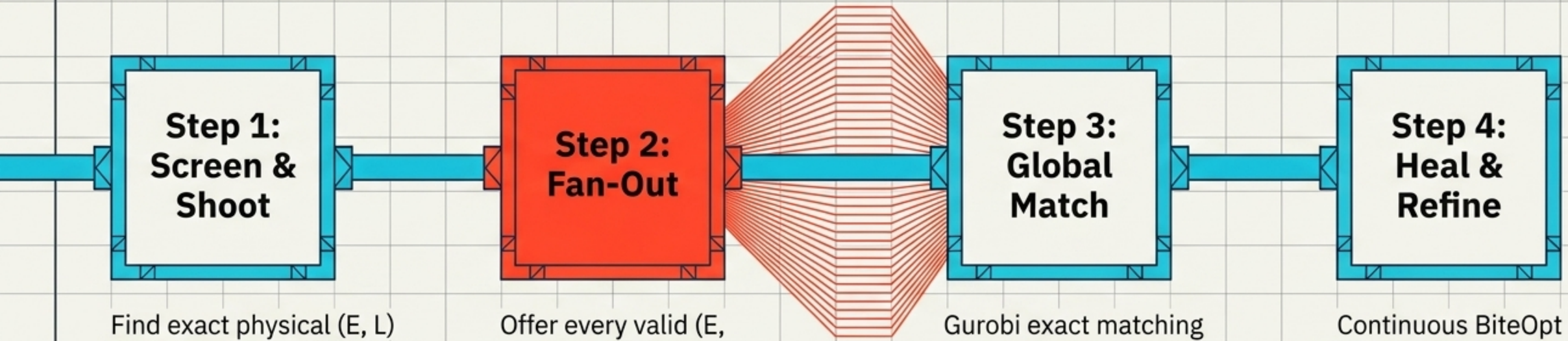
The 400-Destination Fan-Out Pipeline

The Insight: BCP dynamics depend only on the Earth and Moon orbits (E, L). The **destination (D)** only scales the score.



Find exact physical (E, L) transfers using Taylor integration. Separate Direct (Hohmann-like) vs. Weak Stability Boundary (WSB) basins.

Offer every valid (E, L) trajectory to all 400 usable destinations in the dataset.

Gurobi exact matching on the massive fan-out universe.

Continuous BiteOpt refinement targeting the exact Keplerian tolerance proxy.

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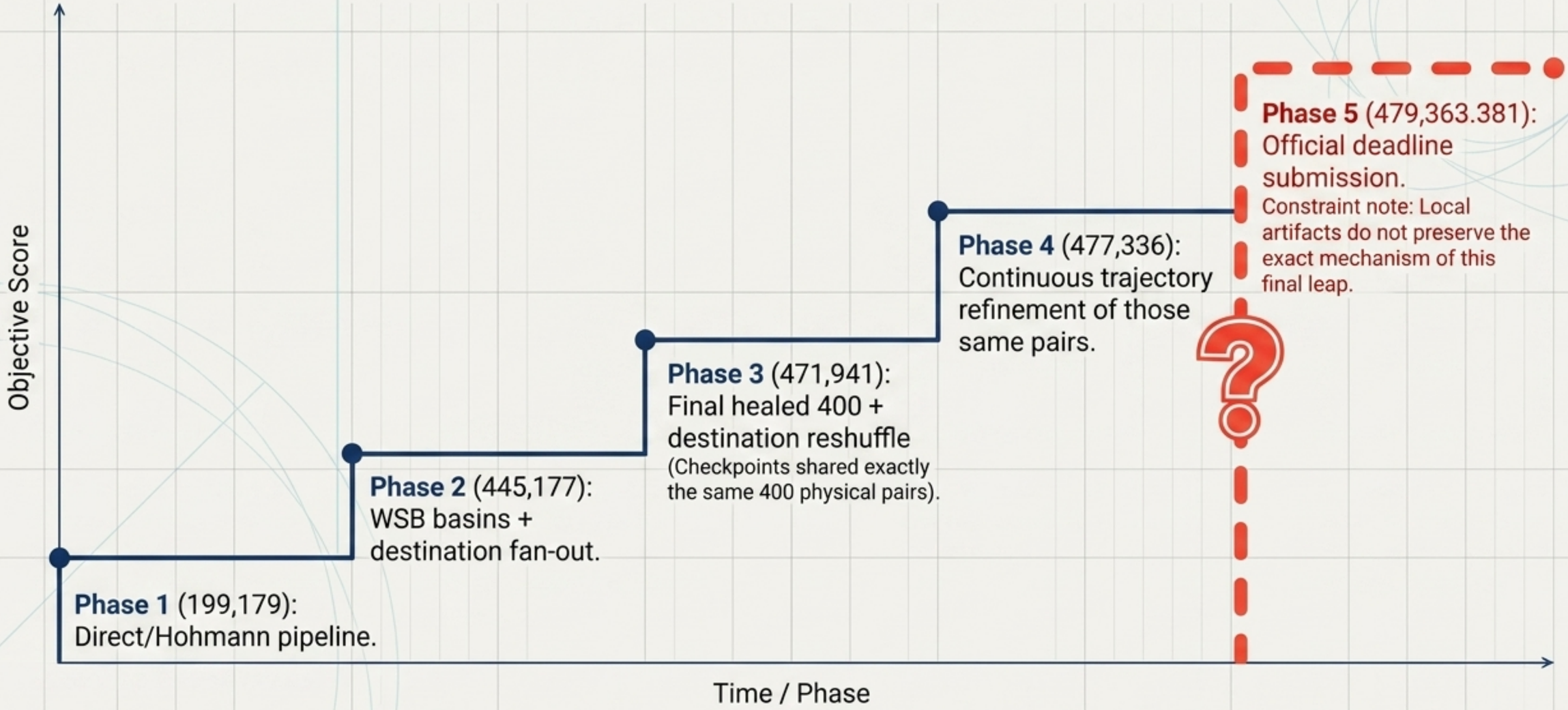
Trajectory Score Lineage (Final Rank 5)



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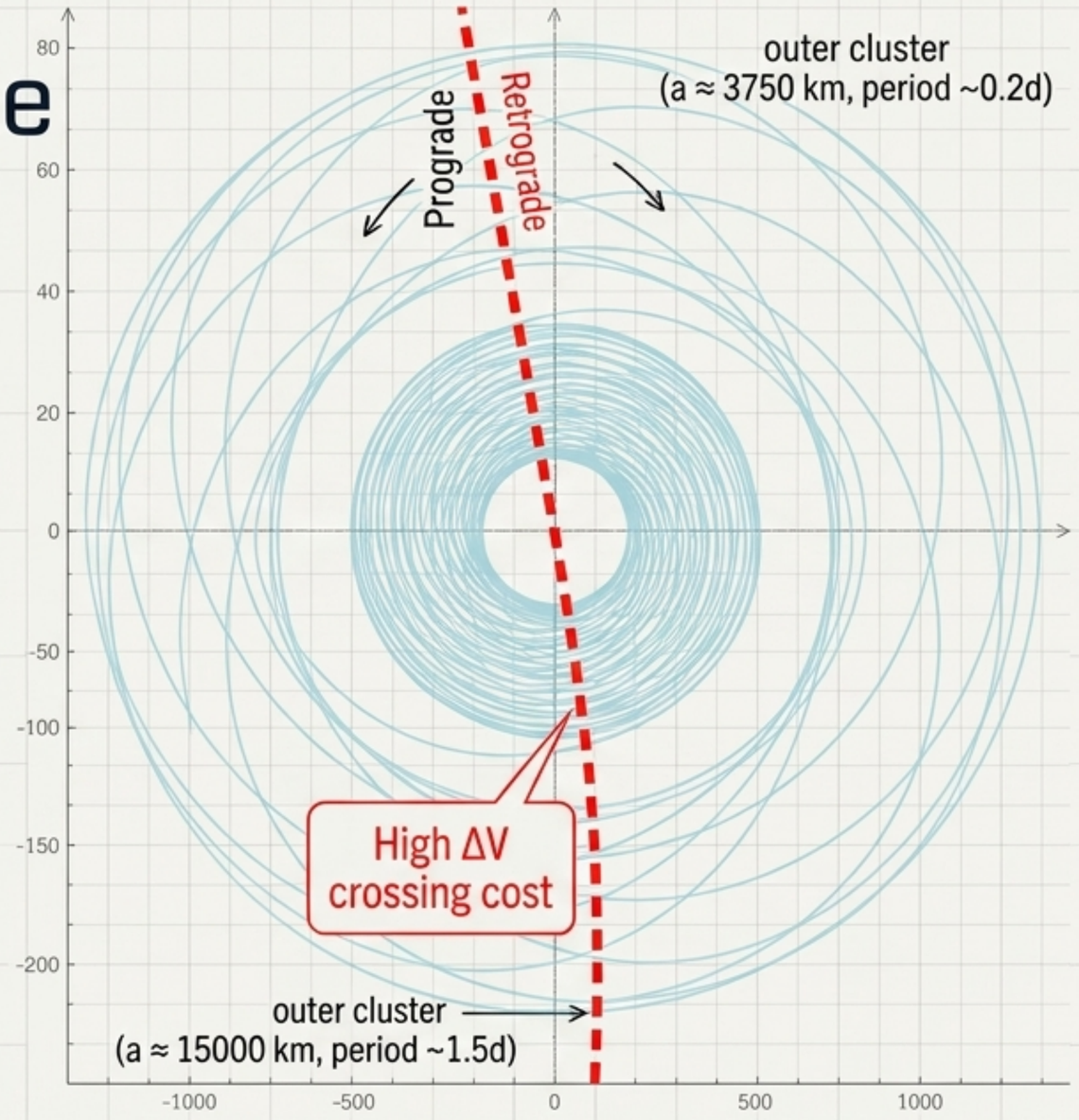
Keplerian TSP: Edges that move with time

The Task: Visit up to 1,051 lunar tomato orbits in minimum mission time.

The Catch: Lambert transfer costs depend heavily on departure epoch and time-of-flight. Waiting is free in ΔV but penalizes the objective clock.

The Constraints: 100 m/s nominal ΔV cap. Exactly 5 exception legs allowed (600 m/s cap).

The Geometric Reality: Low-inclination prograde and retrograde targets are dynamically disconnected. Crossing bands consumes exceptions by force.



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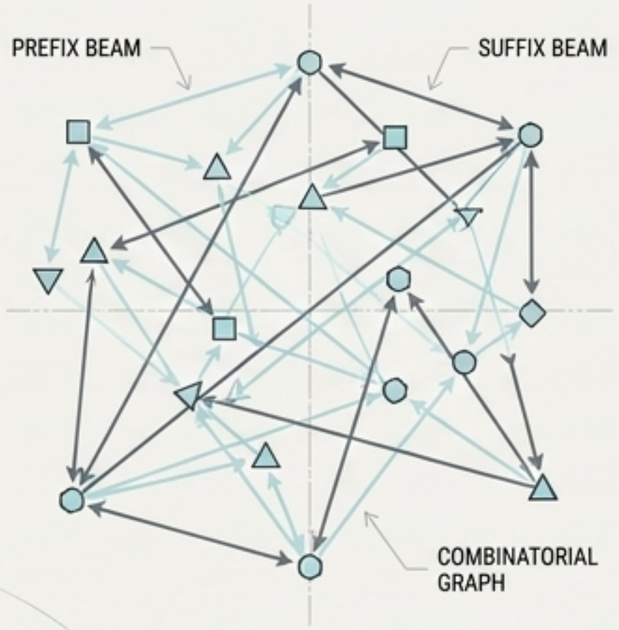
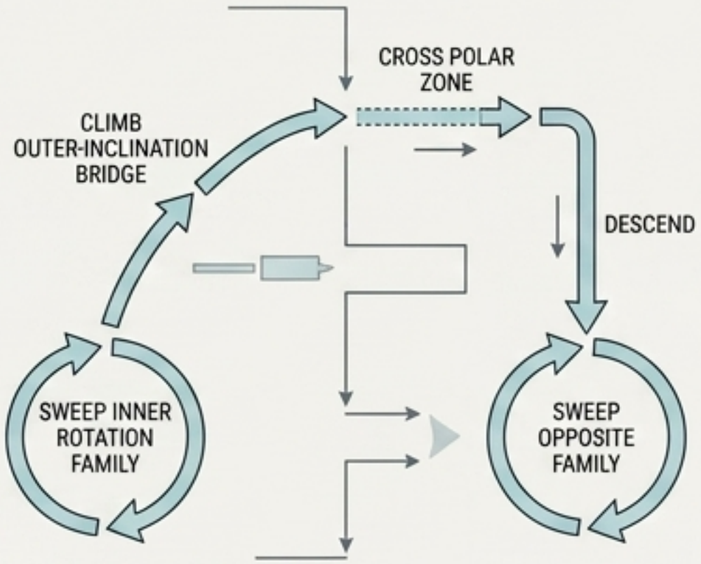
Evolving the search from discrete sequences to continuous geometry

Phase A: Generic TSP (Double-Beam)



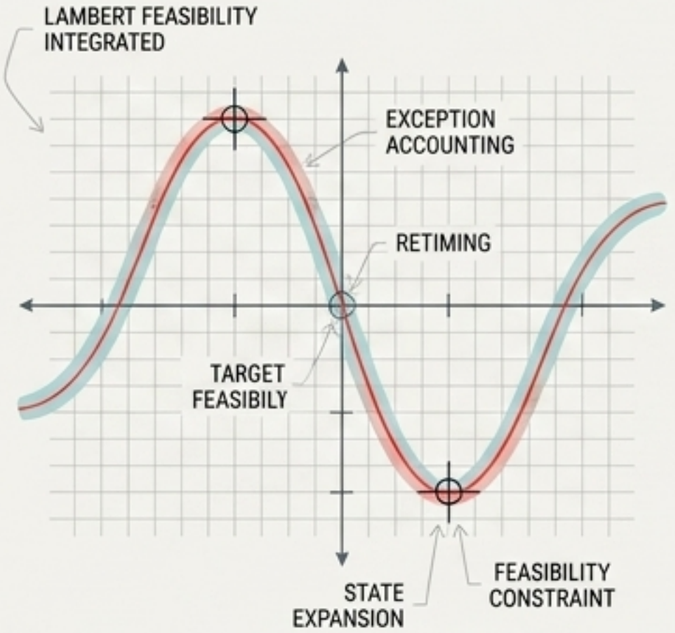
Bidirectional prefix/suffix beams. Failed because it ignored orbital physics.

Phase B: ProRetro Phase Grammar



Segmented the routes: sweep one rotation family → climb an outer-inclination bridge → cross the polar zone → descend → sweep the opposite family.

Phase C: Continuous Lambert Beam



The production engine. Integrated Lambert feasibility, exception accounting, and retiming directly into state expansion. Widths constrained (≈96) to prioritize structure over brute force.

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Three scales, three unique bottlenecks

Small (49 orbits)

Score: 110.878 days
(Rank 3)

The Lesson:

Hit a hard formulation ceiling. Verified across MILP, CP-SAT, and ALNS. Sub-108 day leader scores remain an unexplained anomaly under **zero-wait flight-time floors.**

Medium (181 orbits)

Score: 184.690 days
(Rank 6)

The Lesson:

The unused 5th exception. Forcing the search to spend its final 600 m/s exception on a late high-TOF leg (highlighted in Tomato Red), rather than an early greedy choice, broke a deep local optimum.

Large (1051 orbits)

Score: 436.015 days
(Rank 4)

The Lesson:

Inclination pressure (highlighted in Tomato Red). Ranking candidates by distance to the next scheduled inclination slot drove a 4.3x improvement over early 1888-day baselines.

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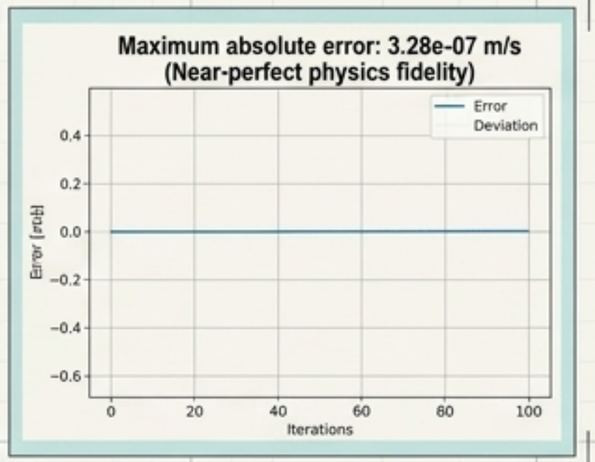
The Numba Lambert Backend: A 113x Search Accelerator

The Problem: The official pykep python-facing path was **too slow** for **inner-loop beam search** evaluating millions of **transfers**.

The Solution: Reimplemented **Izzo's parameterization**, **non-dimensional TOF**, and **Householder iteration** purely in Numba.



The Benchmarks (AMD 9950X 16-core):	
Lambert API (Batch 8):	4,417,722 calls/s (**113.19x speedup** vs serial pykep)
Transfer Table Creation:	Warmed 16-process pool dropped from 24.9s to 1.9s.

Semantics Note: Retained official max_revs [5,2,0] to perfectly mirror ESA validation.

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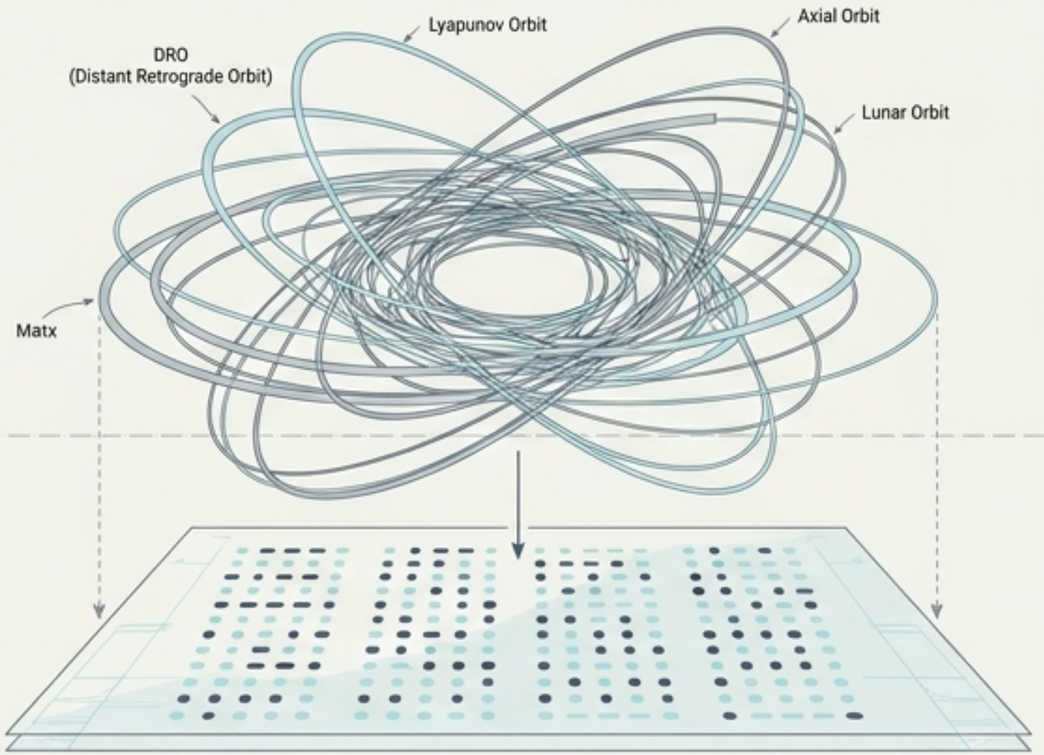
Advertising: Astrodynamics disguised as Sparse Reconstruction

The Task: Minimise spacecraft in lunar orbits so their solar occultations project "_ ___ __ .- - ___ . ..." (Tomatoes for sale) with MSE ≤ 0.05. Max 5 spacecraft per orbit.

The Decisive Reframing: The 779 candidate orbits were fixed. Amplitudes were fixed to 1. **CR3BP dynamics were completely irrelevant** to the solver.

The Real Problem: Sparse signal reconstruction / weighted set cover. **Build a dictionary of (orbit, discrete phase) pulse columns, select the minimum count, and tune phases continuously.**



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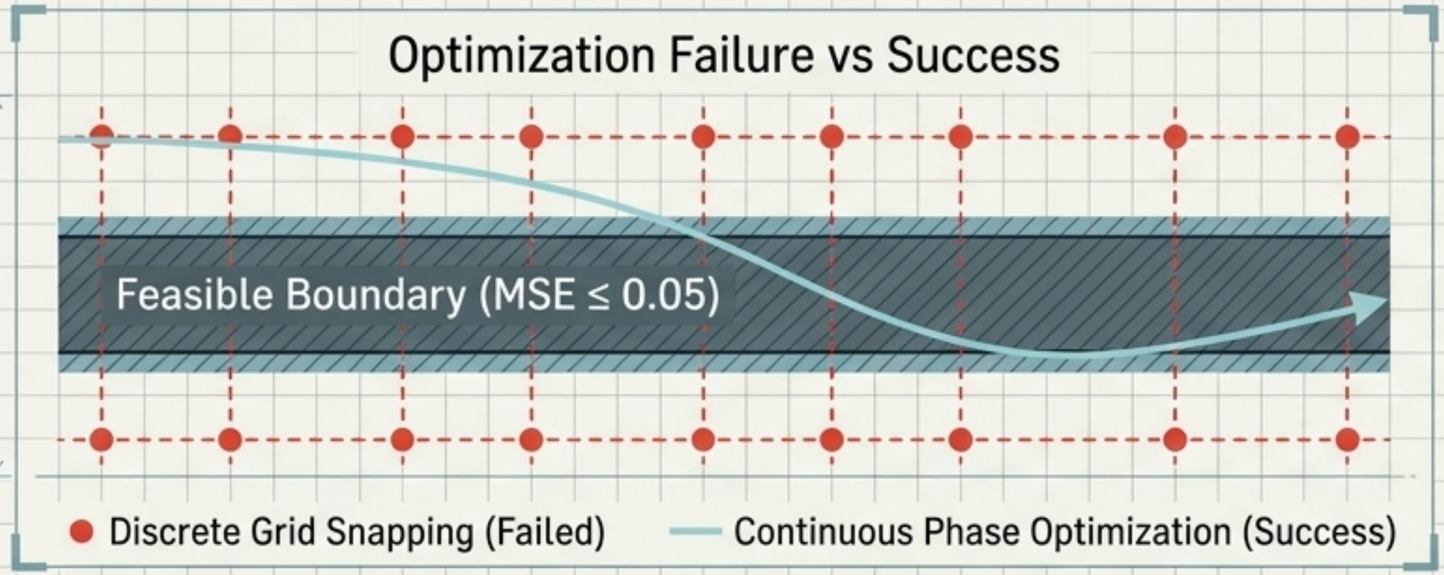
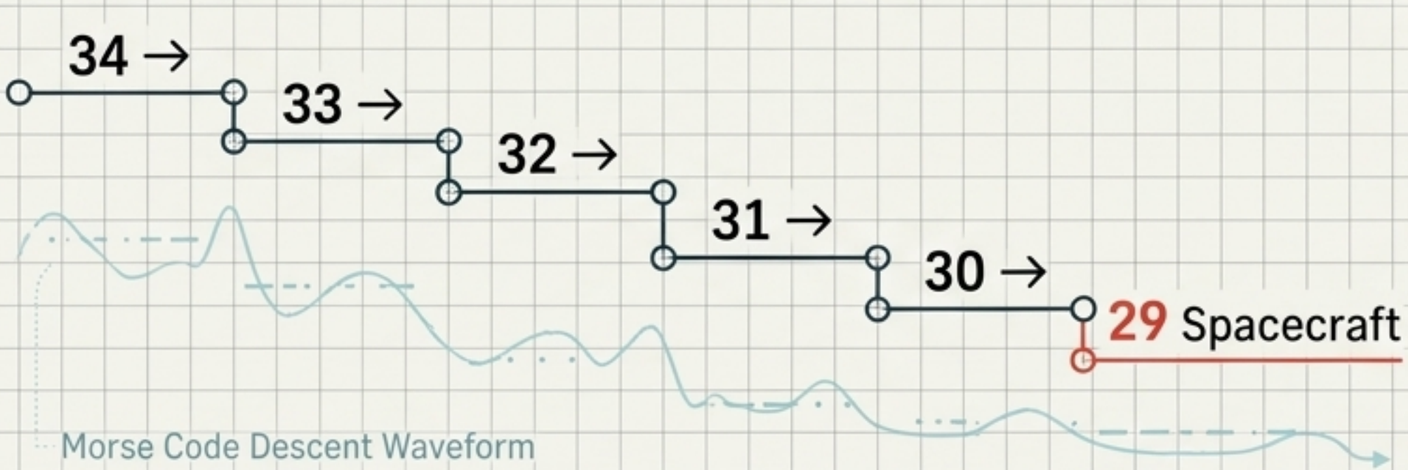
The Discrete Boundary Trap (34 → 29 Spacecraft)





The Trap: Snapping continuous phases onto a discrete dictionary grid resulted in artificial MSEs (e.g., 0.0664, above threshold). Pure ILP stalled at 37-38 spacecraft.

The Fix:
A hybrid continuous-rescue loop. ILP generated a base structure → 16 parallel BiteOpt instances continuously optimized the phases → Powell polish pushed it under the limit.

Result:
29 spacecraft (Rank 2 tie-breaker). Local validations achieved MSEs around 0.0499999999946.

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The Final Pattern: Representation over Algorithm

Challenge	The Wrong Frame	The Right Frame
Matching	3-D matching ILP	**MWIS on the conflict graph** (+856 points instantly)
Logistics	Coupled optimization	**(E,L) Physics ⊥ D Assignment** (Unlocked the 400-destination fan-out)
KTTSP	Euclidean TSP	**Prograde/Retrograde Phase Grammar** (Replaced blind search with structural routing)
Advertising	Astrodynamics	**Sparse Signal Reconstruction** (Removed irrelevant physics entirely)

The Ultimate Lesson: Across all problems, success required changing the representation, validating the exact semantics, and scaling with strict data provenance.

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Appendix A: Authoritative Result Metadata

	Rank	Score	8.000 pts	Data
matching-i	Rank 3	Score: -33,548.085	8.000 pts	Sub: 2026-04-25 22:46 UTC
matching-ii	Rank 3	Score: -73,714.033	10.667 pts	Sub: 2026-04-28 00:28 UTC
trajectory-matching	Rank 5	Score: -479,363.381	10.667 pts	Sub: 2026-06-30 06:00 UTC
tsp-small	Rank 3	110.878421 days	8.000 pts	Sub: 2026-05-28 11:25 UTC
tsp-medium	Rank 6	184.690074 days	6.667 pts	Sub: 2026-06-30 06:00 UTC
tsp-large	Rank 4	436.014924 days	12.444 pts	Sub: 2026-06-30 06:01 UTC
advertising-tie-breaker	Rank 2	29.0 craft	0 pts	Sub: 2026-06-07 02:04 UTC

Note: Final snapshot 20260701T212650Z.json. Submissions were drip-fed; all three late reveals landed within 90 seconds on deadline day.

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Appendix B: Source Discrepancies and Clarifications

Claim/Metric	Stale/Published Value	Authoritative Source Truth
May LaTeX Reports	KTTSP hard (1888.53) and Logistics (471,941)	Radically stale compared to final June breakthroughs (436.015 and 479,363).
BCP Constants	Challenge README printed μ_s = 3.329e5	The executed validator strictly used μ_s = 3.289e5. We optimized against the validator.
KTTSP 6th Element	README labeled it true anomaly (ν)	Executed pykep path treated it as mean anomaly (M).
Advertising Vector	Local candidates retained near 0.04999999999946 MSE	Official snapshot certifies 29 spacecraft count, but cannot confirm exact local phase vector uploaded.

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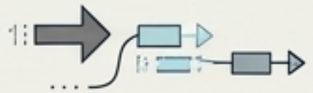
Appendix C: Interventions, Hardware, and Open Questions

Human Interventions that Mattered

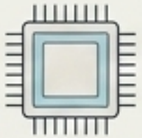
- Use the fifth exception late, not early (Medium: **187d → 184d**).

- Do not trust the discretized ILP result (Advertising: **34 → 29**).

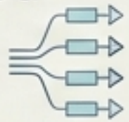


Compute Profile

- Primary production on AMD 9950X 16-core.



- Seed sharding allowed 4-row repair jobs to saturate all 16 cores.



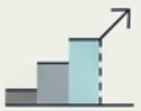
- A planned PR for CHILS upstream will expose AAP limits and partial incumbent seeding.

Open Questions

1. The Easy KTTSP Anomaly (Leader 101.65d vs our zero-wait flight-time floor of 104.3d).

2. Was the leaders' KTTSP edge a time-expanded MILP?

3. The undocumented 477k → 479k final logistics jump.



Architectural scale

vA | 4G, 4G| on previous slides

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